

**skeleton**

25 joints, 3D  
root-relative  
(translation-invariant)

**steerable encoder**

$O(3)$  irreps per joint  
32 type-0 scalars  $s_j$   
8 type-1 vectors  $v_j$  + bone lengths  
 $\Rightarrow$  EQUIVARIANT

**the invariant cut  $\sqcap$** 

parity-even:  $\tanh s_j, \log(1+\|v_j\|), \langle \hat{v}_j, \hat{v}_j' \rangle$   
parity-odd:  $\det[\hat{v}_j^a, \hat{v}_j^b, \hat{v}_j^c]$  (restores chirality)  
bone lengths, anatomical volumes  $\rightarrow 160+123 = 283$ -d

$\Rightarrow$  **INVARIANT for arbitrary weights (Prop. 1)**

**GRU**

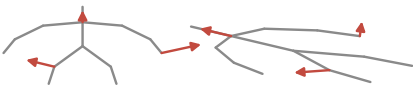
consumes real  
inter-arrival  
times  $\Delta t$

**score**

clinical  
Total Score  
(0-50)

camera A

camera B (rotated)



type-1 features rotate with the camera  
encoder intertwines:  $2.4e-16$

$\sqcap$

**the cut's output is the SAME vector for both cameras**



*bars schematic; the numbers at right are measured*

read-out invariance  
 $8.7e-16$  (fp64 roundoff)

end-to-end score shift  
 $9e-06$  MAD of 50